

Technical Documentation: Interview Integrity Guard

Version: 1.0.0 | **Author:** Senior Frontend Developer | **Subject:** Tab Switch Detection

Project Context

This document details the anti-cheating tab detection system designed to track candidate movements, minimize cheating vectors during online exams, and capture statistics on focus switches.

1. Design Theme & Aesthetics

The dashboard theme is built around a **Futuristic Cyber Hacker Tactical HUD (Heads-Up Display)** style:

- **Obsidian Glass Style:** Deep black background (#020408) with transparent panels utilizing 24px blurring backdrop-filters.
- **Glowing Accents:** Real-time green (Active), red (Inactive), and orange (Pending) glowing rings simulating defense systems.
- **Interactive Matrix Animation:** Cascading digital streams of binary, alphabetic, and Katakana symbols painted on a canvas background in real-time, throttled at 30 FPS.

2. Supported Client APIs & Technologies

This project runs entirely client-side using native HTML5 and Web APIs. It requires **zero external API keys or credentials** to operate.

- **document.visibilitychange:** Fires when the tab goes from visible to hidden (e.g. switching tabs, minimizing browser, launching another app).
- **window.blur:** Fires when the page window loses keyboard focus (e.g. clicking on a secondary screen or DevTools).
- **window.focus:** Fires when the page window regains focus.
- **localStorage:** Used to persist log entries across refresh events.

- **fetch()**: Simulates server-side delivery to the mock api `https://example.com/api/logs` .

3. Debounce State Machine Architecture

To avoid duplicate logs when multiple events fire simultaneously (such as a window blur followed by a visibility change), a centralized state machine with a 1000ms delay aggregates triggers:

1. **Active State:** User is on the page. Glowing green indicator active.
2. **Pending State:** User clicks away or hides page. A 1000ms timer begins. Glowing orange indicator active.
3. **Inactive State:** If the user stays away for >1.0s, the leave log commits. Glowing red indicator active.
4. **Cleared Action:** If they return within <1.0s, the timer is cleared, and no violation is logged.
5. **Paused State:** Disables event tracking. Slate grey indicator active.

4. JSON Log Structure

Logs are written in a standardized JSON format conforming to the anti-cheating tracking guidelines:

Switch-Away (Leave Violation)

```
{
  "event": "tab_switched",
  "tab_switched": true,
  "timestamp": "2026-06-15T14:08:30.124Z",
  "trigger": "Visibility Hidden"
}
```

Returned (Focus Restored)

```
{
  "event": "tab_switched",
  "tab_switched": false,
  "timestamp": "2026-06-15T14:08:32.482Z",
  "trigger": "Visibility Visible"
}
```

5. Project File Map

File Name	Core Responsibility	Key APIs Hooked
index.html	Structure, layout panels, control actions, table wrapper.	DOM Elements, Canvas markup.
styles.css	Glassmorphism layout, glowing animations, hacking theme variables.	CSS Custom properties, Grid.
tabMonitor.js	State machine, events hooks, storage, canvas matrix loop.	VisibilityChange, Blur, Focus, LocalStorage, Fetch, Canvas 2D.
README.md	Setup manual, testing checklists, compatibility lists.	Markdown.

Interview Anti-Cheating & Integrity Detection Console - Confidential Technical Report